



Preparation: Retrieval at saturation



Step 1: Paired reading, one frozen VLM



Step 2: Measure, then look at the evidence



V-NIAH single needle · 1,173 questions · 1 gold page · full recall@16 = 1.000
V-MQAR two-page join · 1,173 questions · 2 gold pages · full recall@16 = 0.998
The reader is shown every gold page in 2,344 of 2,346 samples: what is left to fail is reading.

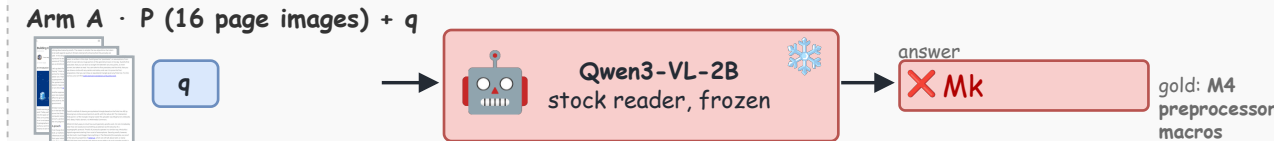
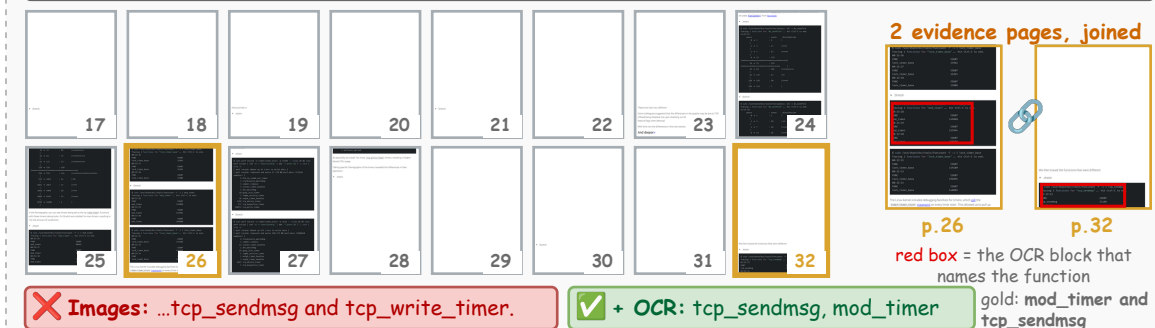
V-NIAH example · Cloudflare blog post, 23 pages (source pp. 1-23) · gold page 12

Q: What specific type of preprocessor macros does the TLS model use for notation?

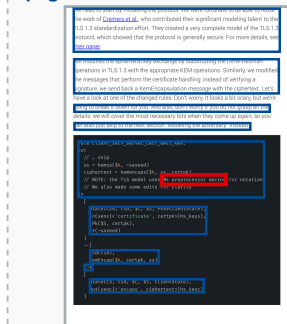


V-MQAR example · Cloudflare blog post, 24 pages (source pp. 17-40) · gold pages 26 + 32

Q: According to the tracing commands on page 26 and page 32, which two specific functions are being monitored?

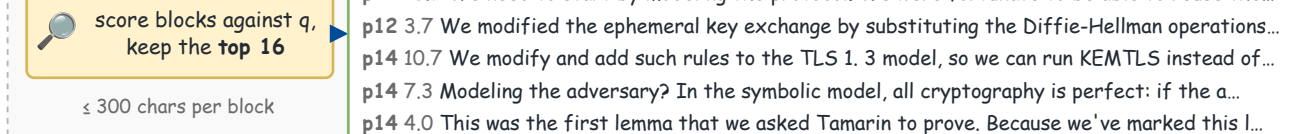


CPU OCR on each of the 16 pages

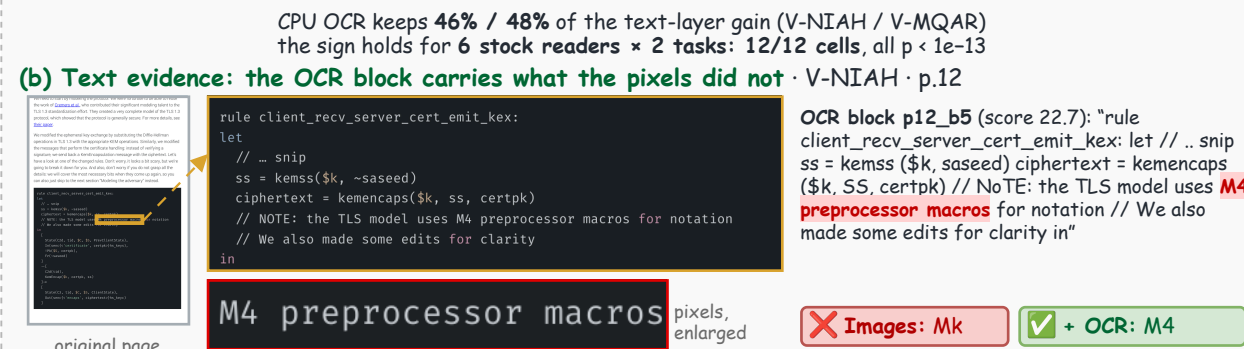
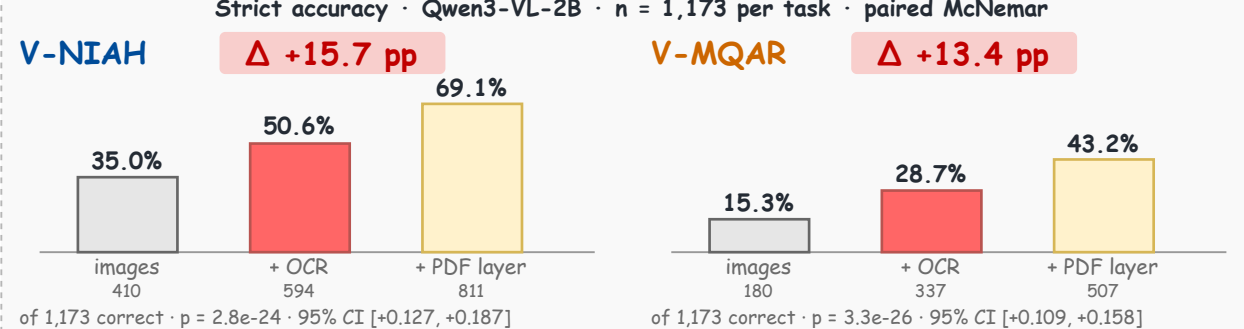


score blocks against q, keep the top 16

≤ 300 chars per block



Same weights, same 16 images, same q: nothing is removed, so Δ is evidence the reader did not use from pixels already in its context. Every document is born-digital, so the PDF text layer is an exact reference; OCR gives up fidelity, not volume, packing +894 chars more than it on 77.1% of samples.



flowchart on the gold page (pixels)

MMLongBench-Doc, single-tag questions: memory - images = +8.5 pp on text (n = 129) vs -4.6 pp on charts / figures (n = 285)
difference +13.1 pp, p = 0.0024: the one external contrast that survives correction

Extracted text amplifies retrieval that works; it does not substitute for it.